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Solar Cycle Prediction: Integrating ARIMA and LSTM Models for Sunspot Dynamics

Abstract:

Sunspots are manifestations of concentrated magnetic flux inhibiting convection. Their cyclical appearance is governed by the approximately 11-year solar cycle. This paper employs advanced time series models to address key solar activity predictions. Three models are established: Model 1: Solar Cycle Period Prediction Model, Model 2: Peak Period Prediction Model, Model 3: Sunspots Number and Area Prediction.

For Model 1, we initially addressed outliers and missing values in the dataset, transforming it into a time series for analysis in the R programming language. We assessed the **autocorrelation function (ACF)**[1], noting elevated values beyond a lag of 120, indicating significant autocorrelation. Slicing the dataset and visualizing sunspot data through **backtesting**[2], we trained the model on a single sample and extended it to all samples. Evaluation through **Root Mean Squared Error (RMSE)**[3] allowed us to predict the onset and conclusion times of the current and next solar cycles.

For Model 2, we commenced with differencing to handle non-stationary data. Utilizing R, we calculated sample autocovariance and autocorrelation coefficients, examining **Autocorrelation Function (ACF)**[4] and **Partial Autocorrelation Function (PACF)**[5] to determine appropriate lag orders (p). The optimal ARIMA(2, 1, 2) model was selected, and **Ljung-Box test results**[6], with X-squared at 15.4, df at 10, and a p-value of 0.12, indicated this indicates that the residuals of the model do not have a significant autocorrelation structure and are consistent with white noise.

For Model 3, an integration of LSTM and ARIMA models was performed. Referencing literature, we derived the relationship between sunspot count and area, simulating the most suitable relationship range across recent solar cycles through time **series slicing**[7]. Validation via **Q-Q Plot**[8] was conducted, and predictions were made for the current and next solar cycles' sunspot count and area.

Ultimately, robustness and sensitivity analyses were executed on the models, with summarized enhancements recommended to improve adaptability across diverse application scenarios.

Key words: LSTM; ARIMA; Sunspots; R; Computer Simulation

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I. Introduction

1.1 Problem Background

Solar activity is a collective term encompassing phenomena such as sunspots, solar flares, and coronal events. Among all solar activities, sunspots represent the most fundamental and conspicuous manifestations. Effectively predicting the number of sunspots proves instrumental in forecasting space weather, ionospheric conditions, and conditions relevant to shortwave radio propagation or satellite communications. The solar magnetic field undergoes polarity changes over the solar cycle, typically lasting around 11 years. However, this cycle is not entirely stable, and the maximum intensity of activity varies over time.

1.2 Restatement of the Problem

Retrieve pertinent literature and address the following three inquiries based on observational data of sunspots:

- Forecast the commencement and conclusion of both the ongoing and subsequent solar cycles.
- Predict the timing of the onset and duration of solar maximum for the forthcoming solar cycle.
- Estimate the quantity and area of sunspots in both the present and future solar cycles, elucidating the model's reliability.

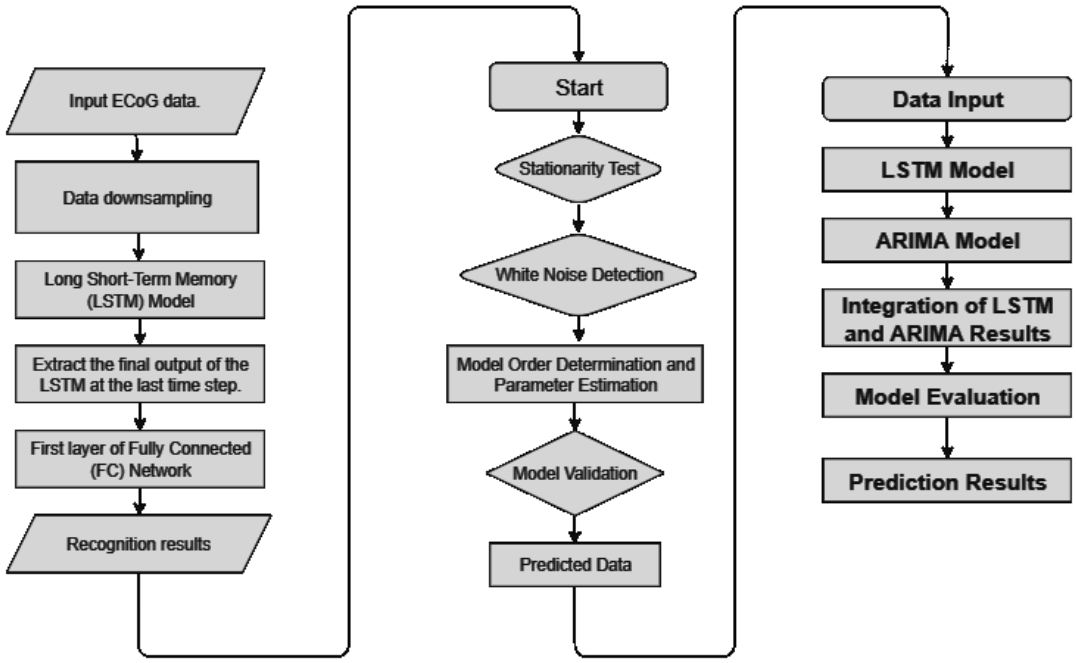
1.3 Literature Overview

A number of researchers have previously contributed to solar cycle prediction model and algorithm.

The study of Xiaohuan Liu et al. provided us with a reference to the underlying rationale ideas and methods for LSTM deep learning method, which predict the amplitude and peak time of solar cycles 25 and 26 by using the monthly relative sunspot number data.[9] A. K. Singh & Asheesh Bhargawa have applied the Hodrick Prescott filtering method to bifurcate each time series into cyclic and trend parts, reconstructed the future time series representing solar cycles 25 and 26.[10] K. M. Hiremath's literature Used physical parameters of the previous solar cycles and by an autoregressive model, predict the amplitude and period of the next 15 solar cycles starting from the solar cycle 23.[11]

1.4 Our Approach

The topic requires reliable solution of predictions related to the solar cycle, and our approach mainly includes the following:



II. Assumptions and Justifications

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

- **Assumption 1**: We assume that the historical patterns observed in solar cycles are indicative of future behavior.

  Justification: This assumption is based on the idea that the Sun's activity tends to follow cyclic patterns, as observed over many solar cycles. While recognizing the inherent variability, we leverage past data to make predictions about the start and end times of the current and next solar cycles.

- **Assumption 2:** We assume that external factors influencing solar activity, such as gravitational interactions or magnetic field dynamics, remain relatively consistent over the short term.

  Justification: This assumption is grounded in the understanding that certain factors contributing to solar cycles exhibit stability over the timescale considered for our predictions. Any major deviations or unprecedented events could challenge the accuracy of our predictions.

- **Assumption 3:** We assume that the relationship between the number and area of sunspots is stable and can be modeled using a combination of time series techniques such as ARIMA and LSTM.

  Justification: This assumption is based on the historical correlation observed between sunspot number and area. We acknowledge the potential influence of

external factors, but we simplify the model by assuming that the primary drivers of sunspot activity can be captured by the selected time series models.

These assumptions provide a foundation for our modeling approach, recognizing the complexities of solar dynamics and the inherent uncertainties associated with long-term predictions. The justifications aim to rationalize the simplifications made for practical modeling while acknowledging the dynamic nature of solar phenomena.

III. Model Preparation

3.1 Notations

Important notations used in this paper are listed in Table 1.

Table1: Notations

Symbol	Description	Unit
t	Time	-
R_t	Sunspot number at time t	-
A_t	Sunspot area at time t	-
SC_t	Solar Cycle at time t	-
SM_t	Solar Maximum at time t	-

The units for sunspot number and area are dimensionless, as they are counts and measures of relative size on the solar surface. The Solar Cycle and Solar Maximum are dimensionless representations of the cyclic nature of solar activity.

3.2 The Data

3.2.1 Data Collection

The data we used mainly include total sunspot number of yearly, monthly, daily and 13-month smoothed monthly, solar cycle predictions on different factors. The data sources are summarized in Table 2.

Table 2: Data source collation

Database Names	Database Websites	Data Type
SIDC	https://www.sidc.be/SILSO/home	Data file
SEPC	www.sepc.ac.cn/SCF_chn.php	Image file
Solar Cycle Science	solarcyclescience.com	Image file
CNKI	https://www.cnki.net	Academic paper

3.2.2 Data Cleaning

From the resources available on the Solar Influences Data Analysis Center (SIDC), sunspot data can be visually observed, the result are shown in Figure14：

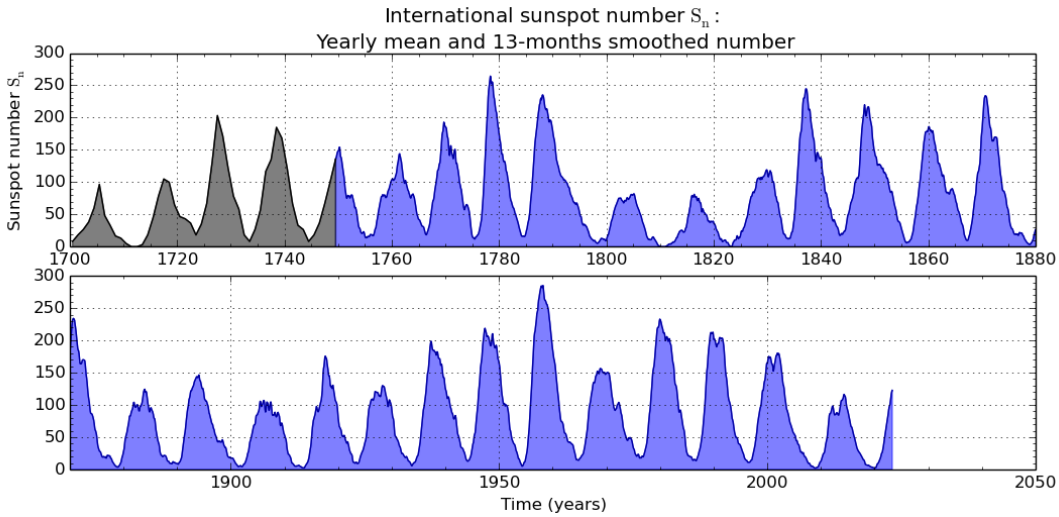


Figure 1: sunspot number

In order to preserve the original data trends while addressing missing values, appropriate normalization and regularization techniques were applied to the data. Long Short-Term Memory (LSTM) was employed for interpolating historical sunspot data. Box plot examination was conducted to identify and address outlier values within the dataset. the result are shown in Figure 2 and 3：

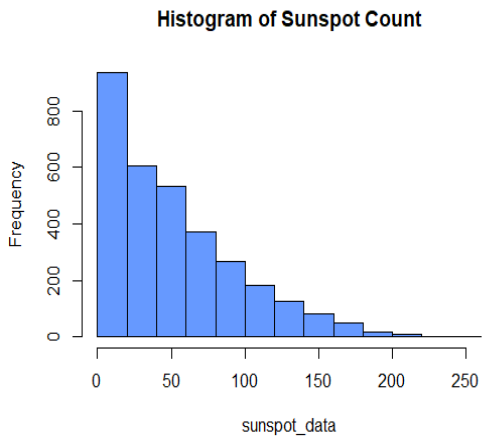


Figure 2: Data after cleaning

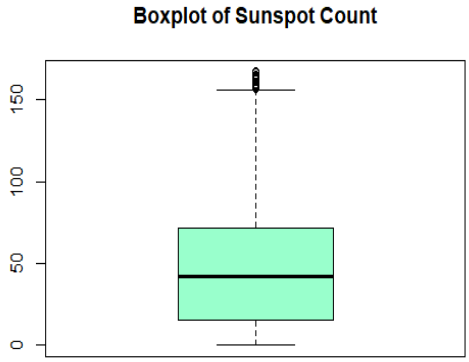


Figure 3: Data after cleaning

IV. Model 1: Solar Cycle Period Prediction Model

The solar cycle denotes the intricate periodic variations in solar activity, typically spanning around 11 years. Through the analysis of historical and forecasted sunspot data, it becomes possible to predict solar cycles. In this study, we utilize sunspot data as the foundational dataset and embark on the development and evaluation of an LSTM model for training. Predictions indicate that the current solar cycle, Cycle 25,

commenced in December 2019 and is anticipated to conclude in January 2030. Subsequently, the next solar cycle, Cycle 26, is projected to commence in January 2030 and conclude in February 2041.

4.1 Description of LSTM Deep Learning Model

The Long Short-Term Memory (LSTM) is a deep learning model specifically designed to address the challenges of handling long sequential dependencies. It represents an extension of Recurrent Neural Network (RNN). The design of LSTM aims to overcome issues such as the vanishing or exploding gradients commonly encountered by traditional RNN models when dealing with long sequences. This enables the model to more effectively capture and learn long-term dependencies.

LSTM incorporates a set of carefully designed gate structures, which aid the model in selectively remembering or forgetting information at different time steps. The core components of LSTM include the forget gate, input gate, cell state, and output gate.The process is depicted in Figure 4.

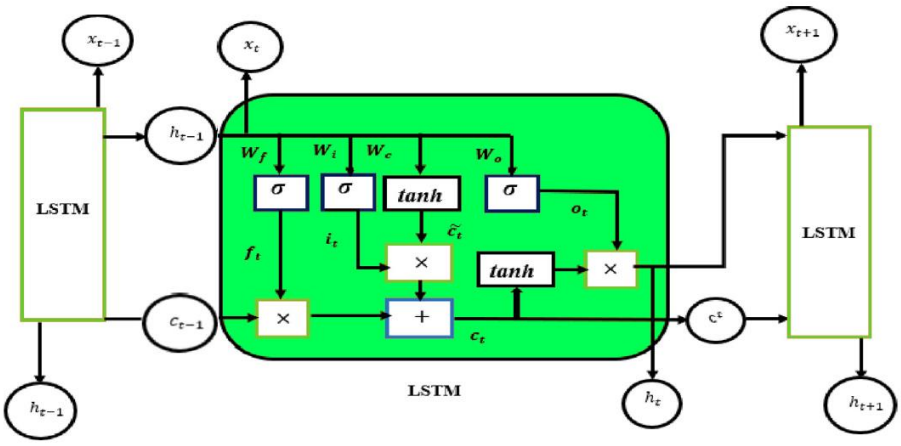


Figure 4: LSTM

4.1.1 Forget Gate

It is responsible for determining how much information to retain in the cell state from the previous time step. It outputs a value between 0 and 1 through the sigmoid activation function, where 0 indicates complete forgetfulness, and 1 signifies complete remembrance.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{1}$$

Here inside the sigmoid function, b_f is the bias and W_f is the weight of the forget gate. h_{t-1} is the output from the preceding cell state, x_t is the input of the present cell state.

4.1.2 Input Gate

It regulates the extent to which new information is incorporated into the cell state. The amount of update to the cell state is determined by the sigmoid activation function, and a new candidate value is generated through the tanh activation function.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{2}$$

$$\tilde{c_t} = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{3}$$

Here inside the sigmoid function, b_i is the bias and W_i is the weight of the input gate. Inside the tanh function, $c\sim_t$ is the candidate cell state, b_c is the bias and W_c is the weight of the cell state.

4.1.3 Cell State

It is updated through a combination of the forget gate and input gate, representing the memory of the LSTM model at different time steps.

$$c_t = f_t \times c_{t-1} + i_t \times \tilde{c_t} \tag{4}$$

c_t combines the information of the preceding cell state being regulated by forget gate and the present updated information.

4.1.4 Output Gate

It determines which portions of the cell state will be output through the sigmoid activation function and scales them using the tanh activation function. This gate decides the hidden state to be output at a specific time step.

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{5}$$

$$h_t = o_t \times \tanh(c_t) \tag{6}$$

Here inside the sigmoid function, W_o is the bias and b_o is the weight of the output gate. o_t is the task of assessing the parts of the cell state that need to be passed on.

4.2 Process of Modeling

4.2.1 Time Series Analysis

The processed data is imported into the R programming language for analysis. The time series is visualized as a 274-year sequence spanning from 1749 to 2023. The result are shown in Figure 5：

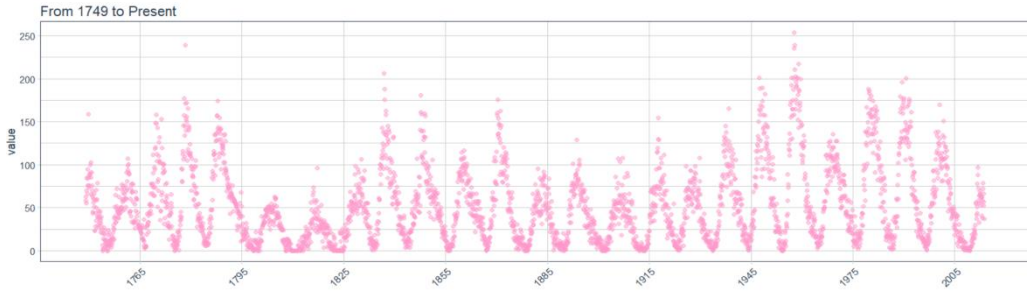


Figure 5: sunspots from 1749 to 2023

4.2.2 ACF Assessment

The Autocorrelation Function (ACF) is a statistical tool used to assess the internal correlation structure of time series data. It measures the self-correlation of a time series at various lag orders, revealing potential periodicities or trends within the data.

$$\rho_k = \frac{\mathrm{Cov}\,(Y_t, Y_{t-k})}{\sqrt{\mathrm{Var}\,(Y_t)\cdot\mathrm{Var}\,(Y_{t-k})}} \tag{7}$$

ρ_k represents the autocorrelation coefficient at a lag of k steps. $Y_t$represents the observed value of the time series at time t. Y_{t-k}represents the observed value of the time series at time t−k. Cov(Y_t,Y_{t-k}) represents the covariance between Y_t and Y_{t-k}. Var(Y_t) and Var(Y_{t-k}) respectively represent the variance of Y_t and Y_{t-k}.

Next, we will employ the Autocorrelation Function (ACF) to assess and generate a 10-year forecast. The ACF is a statistical tool used to evaluate the relationship between a time series and its lagged versions.the result are shown in Figure 6：

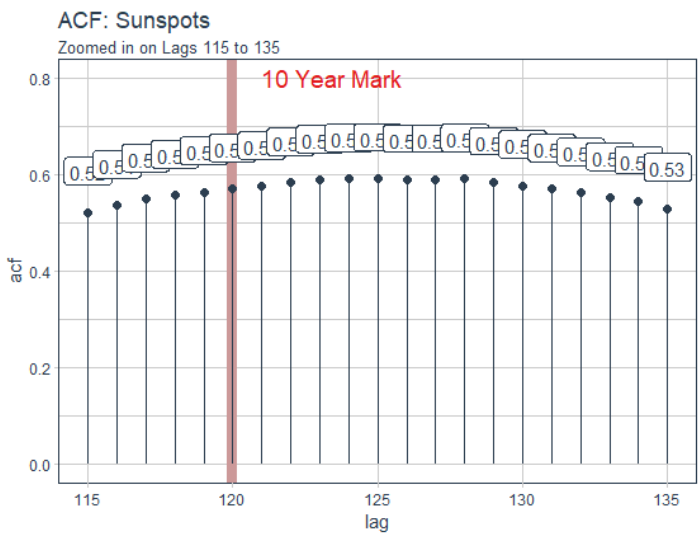


Figure 6: ACF values

4.2.3 Back Test

Cross-validation is the process of developing a model based on a validation set in a subsampled dataset. Its purpose is to determine the expected accuracy level and error range, providing a more realistic simulation of the model's performance on future data, taking into account the temporal sequence.。

We observe the presence of a high-autocorrelation lag extending beyond 10 years, and autocorrelation exceeding 0.5 persists beyond lag 120. Utilizing a filter, the optimal lag is identified at lag 125.

For backtesting, we employ time-series cross-validation. The entire backtesting strategy is depicted in Figure 7, with the results presented as sampling slices.

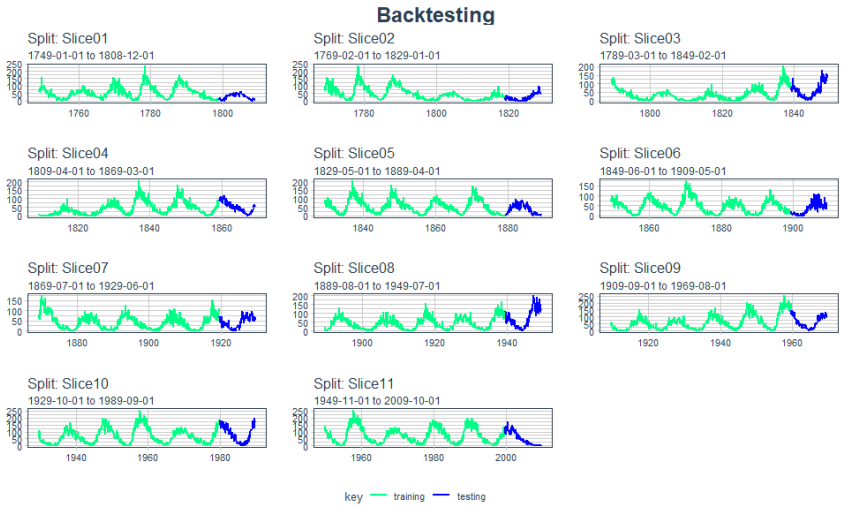


Figure 7: Back testing

4.3 Results

4.3.1 Parameter Estimation

Root Mean Square Error (RMSE) is an evaluation metric used to measure the performance of a predictive model. It is commonly employed to assess the differences between the predicted values of a model and the actual observed values. A smaller RMSE value indicates more accurate predictions, as it measures the root mean square error between the actual observed values and the model's predicted values.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{8}$$

Where n is the number of observations, y_i represents the actual observed values, and $y\hat{}i$ represents the predicted values of the model.

Now, we choose to train the LSTM model on a single sample and then scale it to all samples. For Slice11, we construct the LSTM model and perform RMSE evaluation, the result are shown in Figure 8 and Table 3：

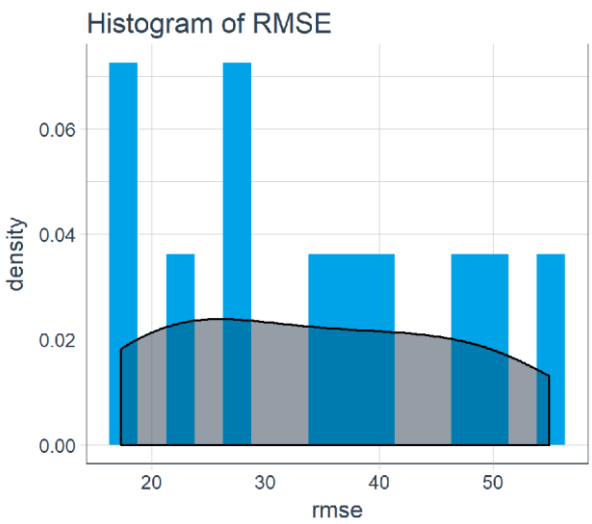


Figure 8: RMSE assessment

Table 3: REMS assessment

Intervals	RMSE
1781-1898	3.08
1807-1908	4.92
1854-1931	5.44
1920-2008	4.82

4.3.2 Calculation Results

Our prediction results are illustrated in Figure 9, and the detailed values are provided in Table 4.

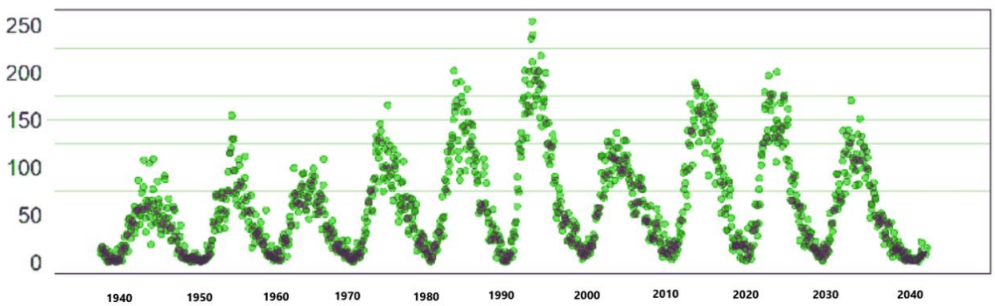


Figure 9: sunspot prediction

Table 4: Solar Cycle Period

Solar Cycle	Start (Y-M)	End (Y-M)
25	2012-12	2030-1
26	2030-1	2041-2

V. Model 2: Peak Period Prediction Model

The highest activity point of sunspots in a cycle is referred to as the solar maximum, while the lowest activity point is known as the solar minimum. We use an ARIMA model to estimate the onset and duration of the peak activity for Solar Cycle 26. The predicted results indicate that the time of onset of the solar maximum for the next solar cycle is January 2035, and the duration of the solar maximum for the next solar cycle is projected to be 57 months.

5.1 Description of ARIMA Time Series Model

Differenced Integrated Moving Average Autoregression (ARIMA) is a statistical model commonly employed in time series analysis and forecasting. It incorporates three components: Autoregressive (AR), Integrated (I), and Moving Average (MA), designed to handle non-stationary time series data and capture trends and periodicity within the data. The notation for an ARIMA model is ARIMA(p, d, q), where p denotes the order of the autoregressive part, d represents the order of differencing, and q signifies the order of the moving average part. The model workflow is illustrated in Figure 11.

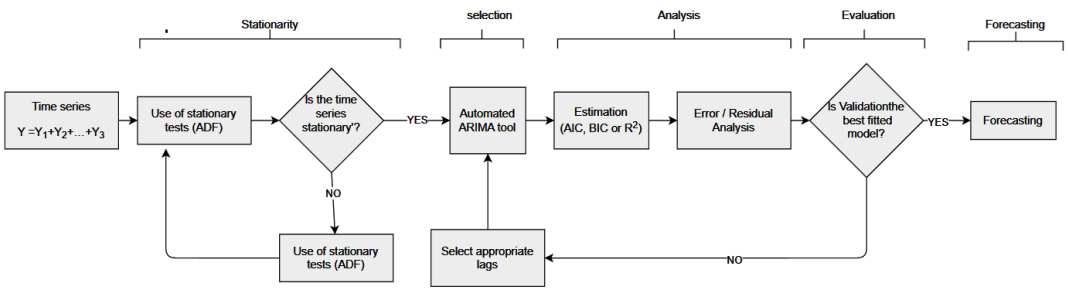


Figure 11: ARIMA Model

5.1.1 Autoregressive (AR) Part

Autoregression refers to the linear relationship between the current observation and past observations at several time steps. The autoregressive component is denoted by p, where p is the order of autoregression, representing the number of past time steps considered.

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t \qquad (1)$$

Here, Y_t represents the observed value of the time series at time t. c is the intercept term.ϕ_1, ϕ_2, …, ϕ_p are the lag parameters of the model, indicating the linear relationship between the current observation and past p observations. ε_t is the white noise error term, representing the random fluctuations that the model has not captured.

5.1.2 Differential (I) Part

Differencing involves the operation of taking differences on a time series, aiming to make the data stationary. Stationary time series are easier to model and predict. The parameter d represents the order of differencing, indicating the number of times differencing is performed.

$$\text{First Difference: } \Delta Y_t = Y_t - Y_{t-1} \quad (2)$$
$$\text{Second Difference: } \Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1} \quad (3)$$

Here, Y_t represents the observed value of the time series at time t, ΔY_t denotes the first-order difference of Y_t, and $\Delta^2 Y_t$ represents the second-order difference of Y_t.

5.1.3 Moving Average (MA) Part

Moving Average MA represents the linear relationship between the current observation and past white noise errors at several time steps. The MA component is denoted by q, where q is the order of the moving average, indicating the number of past white noise errors considered.

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q} \quad (4)$$

Where, μ is the mean, ε_t, ε_{t-1} ,…, $\varepsilon_{t-q}$are the white noise error terms. θ_1, θ_2, …, $\theta_q$are the parameters of the moving average part, representing the linear relationship between the current observation and past white noise errors.

5.2 Process of Modeling

5.2.1 Difference Calculation

To ensure the stationarity of time series data, differencing is applied to non-stationary data, the result is shown in Figure 11.

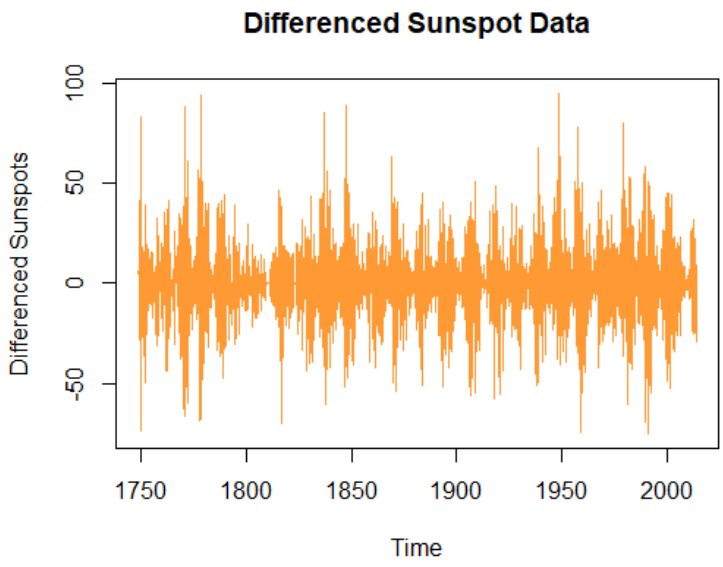


Figure 11: Differenced Sunspot Data

5.2.2 Model fitting: ACF and PACF

First, we compute the sample autocovariance coefficients in the R language,

$$\hat{\gamma}(h) = n^{-1} \sum_{t=1}^{n-h} (x_{t+h} - \bar{x})(x_t - \bar{x}) \tag{5}$$

For h=0,1,...,n−1, we have γ^(−h)=γ^(h). Here, n is the total number of samples,h is the lag order, and $\bar{x}$ is the sample mean. Next, we compute the sample autocorrelation coefficients.

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)} \tag{6}$$

We determine the appropriate lag order \(p \) by observing the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF), as illustrated in Figure 12 and 13.

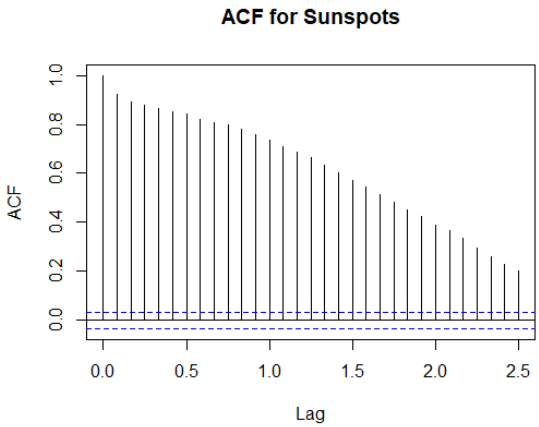


Figure 12: ACF Model

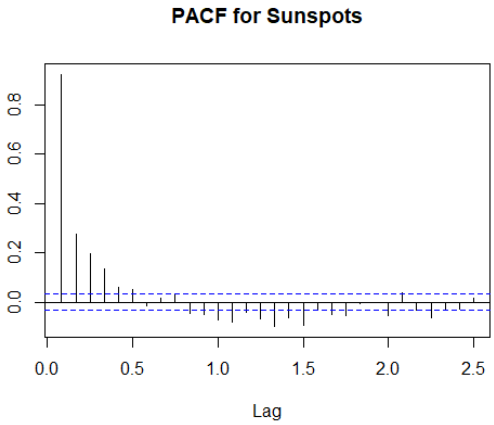


Figure 13: PACF Model

Based on the output results, we calculate all permutations within 2 orders and determine that the optimal model order is (p, d, q) = (2, 1, 2). Therefore, we employ the ARIMA(2, 1, 2) model, and the results are presented in Table 5.

Table 5: ARIMA Output

	Coefficients	s.e
AR1	1.3307	0.0289
AR2	-0.3807	0.0272
MA1	-1.7582	0.0198
MA2	0.7979	0.0186

5.2.3 Model Diagnostics

By fitting the model and forecasting the observed data, a predicted sequence can be obtained. The residual sequence is derived by subtracting the corresponding predicted

values from the observed values. Residuals reflect the unexplained portion of the model, i.e., the random fluctuations or other unknown factors that the model has failed to capture. The results is shown in Figure 14.

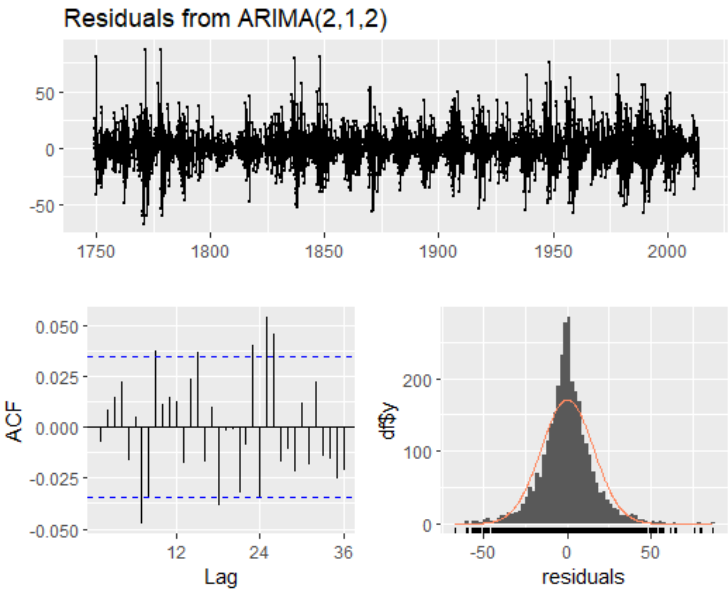


Figure 14: Residuals for ARIMA

5.3 Results

5.3.1 Parameter Estimation

Ljung-Box test is a statistical method used to check whether there is autocorrelation in the residuals of a time series. This test is based on the chi-square distribution. The null hypothesis of the test is that there is no autocorrelation in the residual sequence, meaning that the residuals are independently and identically distributed white noise. The alternative hypothesis is that there is autocorrelation in the residual sequence.

$$Q = n(n+2)\sum_{k=1}^{h} \frac{\hat{\rho}_k^2}{n-k} \tag{7}$$

Where, n is the sample size, h is the lag order, and $\hat{\rho}_k$ is the autocorrelation coefficient of the residual sequence.

We obtain the results: X-squared is 15.4, df is 10, p-value is 0.12. This indicates that the residuals of the model do not exhibit a significant autocorrelation structure and are consistent with the characteristics of white noise.

5.3.2 Calculation Results

We present the results of the ARIMA model and the previously calculated results of the LSTM model in Table 6.

Table 6: Maximum Prediction

Solar Cycle	Maximum(Y-M)	Duration(Days)	Peak

26	2035-1	156	135.0

VI. Model 3: Sunspots Number and Area Prediction

Sunspots are regions of reduced surface temperature caused by concentrations of magnetic flux inhibiting convection. Based on the predicted results of Solar Cycle 25 and Solar Cycle 26 obtained in the previous sections, we can determine the number of sunspots and establish the relationship between sunspot count and area. This relationship is modeled using a combination of LSTM and ARIMA, the number of sunspots in the current solar cycle is 2148, area of sunspots in the current solar cycle is 157.4, area of sunspots in the current solar cycle is 2433, area of sunspots in the next solar cycle is 169.2.

6.1 Description of Combination Model of LSTM and ARIMA

6.1.1 LSTM PART

We use LSTM to capture long-term dependencies, and the output of LSTM can be expressed as:

$$Y_t^{LSTM} = f_{LSTM}(X_t^{LSTM}, \theta_{LSTM}) \tag{1}$$

Where Y_t^{LSTM} is the predicted output of the LSTM model at time t, X_t^{LSTM} is the input to the LSTM model, and θ_{LSTM} represents the parameters of the LSTM model.

6.1.2 ARIMA PART

We use the ARIMA model to handle short-term trends in the time series, which can be expressed as:

$$Y_t^{ARIMA} = f_{ARIMA}\left(Y_{t-1}, Y_{t-2}, \dots, Y_{t-p}; \theta_{ARIMA}\right) \tag{2}$$

Where, Y_t^{ARIMA} is the predicted output of the ARIMA model at time t, Y_{t-1}, Y_{t-2}, …, Y_{t-p} are the lag terms of the ARIMA model, and θ_{ARIMA} represents the parameters of the ARIMA model.

6.1.3 Combination

To combine the outputs of LSTM and ARIMA, a linear weighting approach is adopted to form the final prediction result:

$$Y_t^{Combined} = \alpha \cdot Y_t^{LSTM} + (1-\alpha) \cdot Y_t^{ARIMA} \tag{3}$$

Where α is the weight coefficient used to balance the contributions of the two models.

6.2 Process of Modeling

We obtained Figure 15 from the Peter Meadows website， yellow for groups smaller than 100 millionths, green for groups between 100 millionths and 500 millionths, purple for groups between 500 millionths and 1000 millionths and red for groups greater than 1000 millionths， If the area of a group is very small (for example, for groups of A and B types), then its area is considered zero. The occurrence of the highest number of high-area groups typically happens one to two years before and after the maximum of sunspots.The units of sunspot area is millionths of the Sun's visible hemisphere.

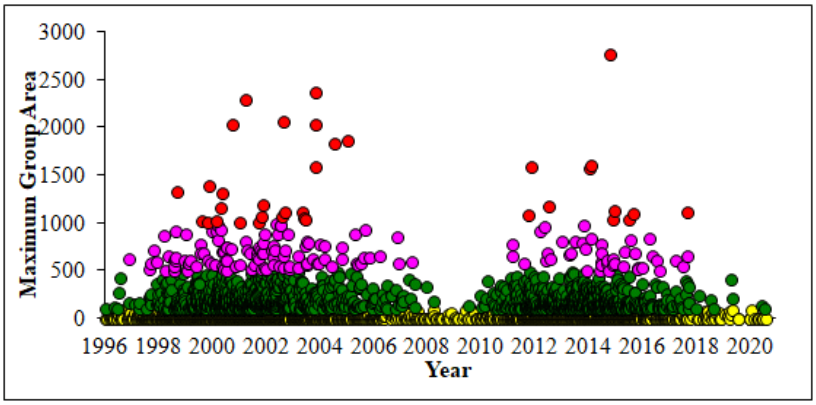


Figure 15: sunspot group size

6.2.1 Sunspot Calculation

The Sunspot Mask

The most accurate calculation method. Sunspot area is determined by creating a pixel mask (i.e., including all pixels comprising a sunspot), followed by the calculation of the pre-scaled area of each pixel within the mask.

Mathematical and Graphical Method

$$\mathbf{A}_{\mathrm{M}} = \frac{\mathbf{A}_S 10^6}{2\pi \mathbf{R}^2 \cos(\mathrm{p})} \approx \frac{\mathbf{A}_S 10^6}{2\pi \mathbf{R}^2 \cos(\mathrm{B}) \cos(\mathrm{L} - \mathbf{L}_0)} \qquad (4)$$

Where, A_M = sunspot area in millionths of the Sun's visible hemisphere, ρ = the angular distance on the surface of the Sun from the center of the disk to the group, A_S = measured sunspot area (e.g. square mm or square inch), R = radius of solar drawing (e.g. mm or inches), B = heliographic latitude of sunspot group (degrees), L = heliographic longitude of sunspot group (degrees) and L_0 = heliographic longitude of the center of the disk (degrees).

We obtained the following relationships through numerical simulations:

Table 7: Sunspot area versus annual average relationship.

Intervals	Relationship
1875-2004	A = (13.1 ± 3.5)R
1875-1976	A = (14.1 ± 3.2)R
1977-2004	A = (9.8 ± 2.1)R

6.3 Results

6.3.1 Parameter Estimation

Quantile-Quantile (Q-Q) Plot is a graphical tool used to assess whether a sample comes from a particular theoretical distribution. In time series analysis and statistics, Q-Q plots are often employed to evaluate whether the residual sequence conforms to a normal distribution.

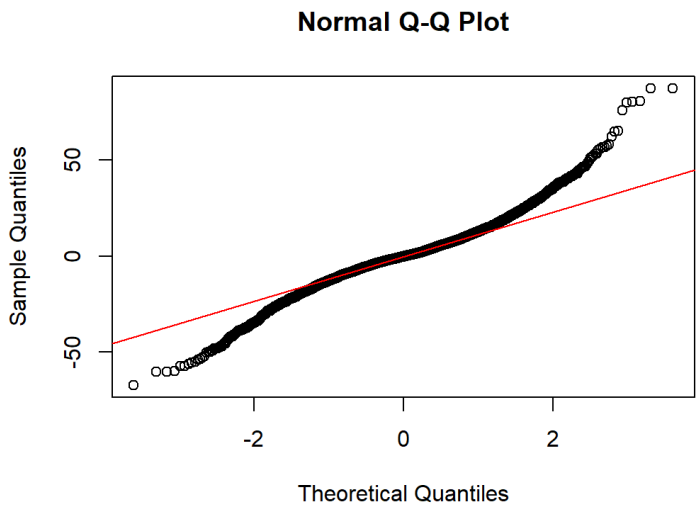


Figure 16: Normal Q-Q Plot

The scatter plot approximately aligns along a straight line, with some areas showing curvature or deviation. Hence, we can infer that the model roughly conforms to the theoretical distribution.

6.3.2 Calculation Results

Our final results are summarized in Table 8.

Table 8: Sunspot Numbers and Areas Prediction

Solar Cycle	Numbers	Areas
25	2148	157.4
26	2433	169.2

VII. Test the Model

7.1 Sensitive Analysis

7.1.1 Sensitivity to Model Parameters:

- ARIMA Order Selection:

We evaluate the impact of different ARIMA model orders on the prediction results by changing parameters such as autoregressive (p), differencing (d), and moving average (q) orders. The optimal orders and model are identified, as shown in Figure 14.

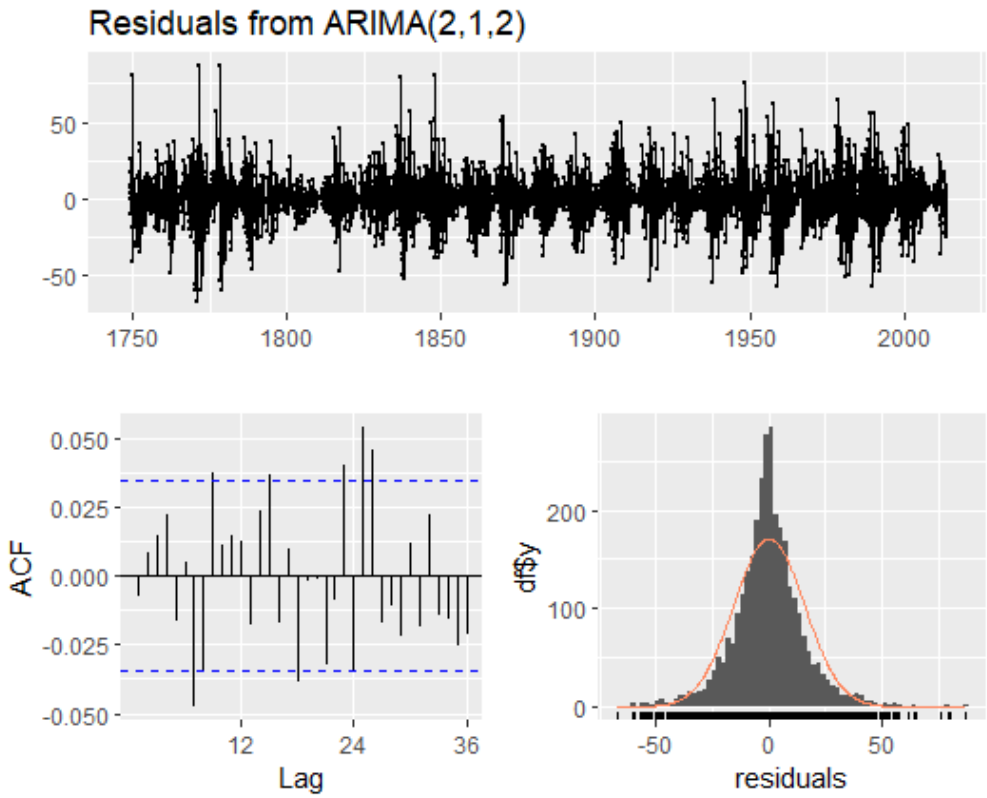


Figure 14: Residuals for ARIMA

- LSTM Hyperparameter Tuning:

We conduct sensitivity analysis by adjusting hyperparameters in the LSTM model, including the number of layers, units, and learning rate, to evaluate the model's responsiveness.

Sensitivity to Data Inputs:

- Varying Historical Periods:
  We consider slicing with different lengths of training and testing data to explore how the model responds to variations in historical periods. Refer to Figure 7 for details.

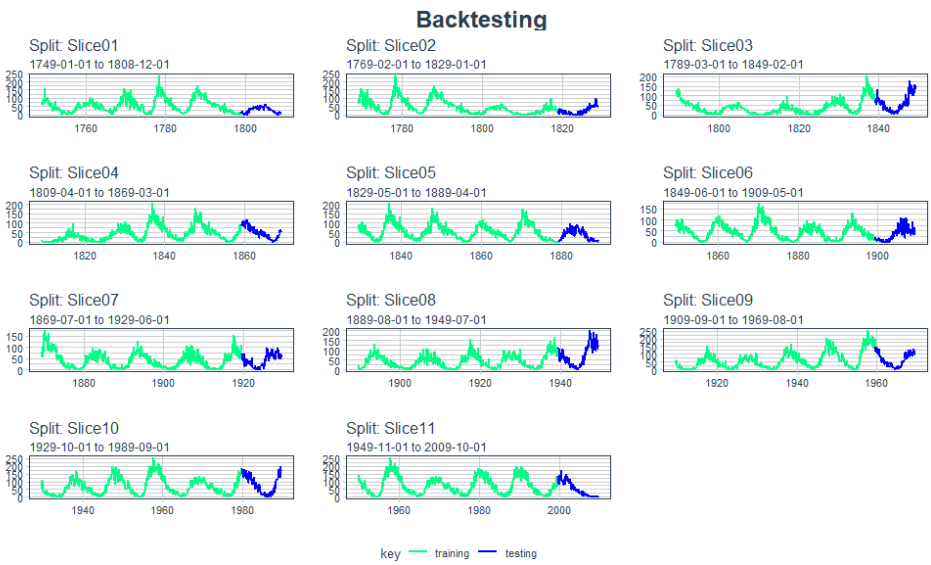
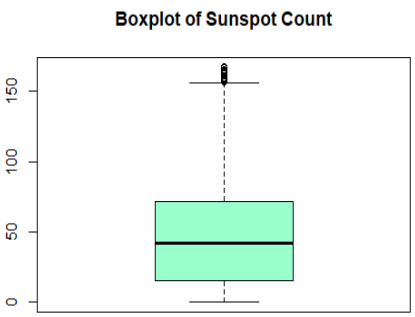


Figure 7: Back testing

- Handling Missing Data:

We investigate the sensitivity of the model to varying degrees of data missingness by simulating different levels of data absence. Refer to Figure 2 and Figure 3 for details.



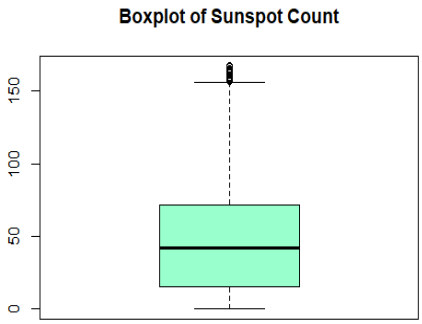


Figure 2: Data after cleaning

Figure 3: Data after cleaning

VIII. Conclusion

8.1 Summary of Results

8.1.1 Result of Problem 1

According to the calculation results from Model 1, the results are shown in Table 4.

Table 4: Solar Cycle Period

Solar Cycle	Start (Y-M)	End (Y-M)

25	2012-12	2030-1
26	2030-1	2041-2

8.1.2 Result of Problem 2

According to the calculation results from Model 2, the results are shown in Table 6.

Table 6: Maximum Prediction

Solar Cycle	Maximum(Y-M)	Duration(Days)	Peak
26	2035-1	156	135.0

8.1.3 Result of Problem 3

According to the calculation results from Model 3, the results are shown in Table 4.

Table 8: Sunspot Numbers and Areas Prediction

Solar Cycle	Numbers	Areas
25	2148	157.4
26	2433	169.2

8.2 Strength and Weakness

Strengths:

- **Integration of ARIMA and LSTM:**
  The combination of ARIMA and LSTM models allows us to capture both short-term and long-term dependencies in the time series data. ARIMA handles linear trends and seasonality, while LSTM excels in capturing complex non-linear patterns.
- **Incorporation of Historical Patterns:**
  Leveraging historical sunspot data provides a foundation for understanding cyclic patterns and improving the accuracy of predictions.

Weaknesses:

- **Sensitivity to Assumptions:**
  The reliability of the model depends on the validity of the assumptions made, such as the stability of external factors influencing solar activity.
- **Uncertainties in Solar Physics:**
  Inherent uncertainties in solar physics and the complex interplay of various factors introduce challenges in making precise predictions.
- **Data Limitations:**
  The quality and availability of historical data may impact the accuracy of predictions, especially when dealing with events outside the historical range.

8.3 Possible Improvements

- **Enhanced External Factor Consideration:**

Further research could explore the incorporation of additional external factors, such as variations in the solar magnetic field or interactions with other celestial bodies, to enhance model accuracy.

- **Advanced Machine Learning Techniques:**
  Investigation into more advanced machine learning techniques beyond LSTM, such as transformer models, may offer improvements in capturing complex temporal patterns.
- **Real-time Data Integration:**
  Integrating real-time observational data into the model could enhance its adaptability to unforeseen events and improve accuracy in predicting solar cycle characteristics.

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